

Lecture 14: Responsible Agents and the Agentic Future

Safety, scientific integrity, governance, and plausible futures

FYS4411 — Reinforcement Learning and Agentic AI
University of Oslo, Department of Physics

2026

Opening proposition: agency is an amplifier

A language model generates candidate continuations. An agent can convert a continuation into a sequence of consequential actions: query a database, modify code, schedule an experiment, communicate externally, or delegate to other agents.

One useful abstraction is

$$\text{impact} \propto \underbrace{\text{capability}}_{\text{what can be inferred}} \times \underbrace{\text{access}}_{\text{what can be reached}} \times \underbrace{\text{autonomy}}_{\text{what can be decided}} \times \underbrace{\text{persistence}}_{\text{how long it continues}} \times \underbrace{\text{scale}}_{\text{how often it is replicated}} .$$

The same factors that make agents valuable also magnify error, misuse, and loss of control. Responsibility is therefore a property of the complete sociotechnical system.

Learning goals

After this lecture, you should be able to

- ▶ distinguish model-level limitations from risks created by tools, memory, orchestration, and deployment context;
- ▶ threat-model an agent trajectory, including prompt injection, privilege escalation, long-horizon drift, and multi-agent failure;
- ▶ design layered controls: least privilege, sandboxing, approval gates, provenance, monitoring, evaluation, and incident response;
- ▶ formulate scientific validation requirements that remain external to the proposing agent;
- ▶ evaluate claims about future agents by separating measurements, extrapolations, assumptions, and value judgements.

The central dilemma

The promise

Agents can search larger hypothesis spaces, integrate heterogeneous evidence, operate instruments, execute simulations, and preserve a detailed computational record.

The danger

Agents can also act on false beliefs, follow hostile instructions, reproduce a flawed method at scale, obscure accountability, or optimize a measurable proxy instead of the scientific objective.

The responsible question is not “Is the model intelligent?” It is:

What evidence justifies granting this system this authority, in this environment, for this duration?

Part	Central question
I. Responsibility	What changes when a model becomes an acting system?
II. Failure and misuse	How can agents fail accidentally, adversarially, or systemically?
III. Engineering controls	Which safeguards constrain authority and produce evidence?
IV. The agentic future	Which trends are measured, and which futures remain conditional?
V. Synthesis	What professional obligations follow for scientists and engineers?

Motivated by Agents for Science curriculum: human–AI workflows, evaluation, failures, safety, novelty, and agent training.

Part I: What responsible agency means

Safety is a systems property, not a model personality

A responsible agent is constrained and accountable

Consider an agent as a tuple

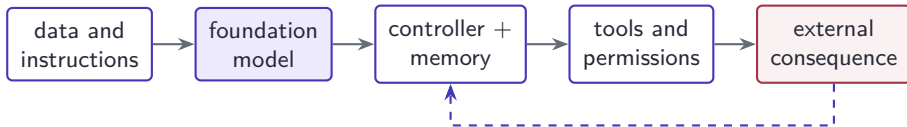
$$\mathcal{G} = \langle \pi_{\theta}, S, A, T, M, B, H \rangle,$$

where π_{θ} is a model-guided policy, S the state, A admissible actions, T tools, M memory, B resource budgets, and H human or institutional oversight.

Responsible operation requires that

- ▶ authority is bounded by explicit permissions and duration;
- ▶ consequential claims and actions are traceable to evidence;
- ▶ violations are detectable, interruptible, and reviewable;
- ▶ a named institution or person remains accountable for deployment decisions.

Model responsibility versus system responsibility



Model question

Does the model produce unsafe, deceptive, biased, or unreliable outputs?

System question

Can those outputs obtain authority, persist, evade detection, or cause irreversible effects?

A comparatively capable model in a read-only sandbox may be safer than a weaker model with broad credentials and no monitoring.

A single response is an incomplete unit of analysis. Let

$$\tau = (s_0, a_0, o_1, s_1, \dots, a_{T-1}, o_T, s_T)$$

denote the full trajectory. For hazards h_i with severities S_i , a simple expected-loss model is

$$\mathbb{E}[L \mid \tau] = \sum_i P(h_i \mid \tau) S_i.$$

- ▶ The same action can be harmless in one state and dangerous in another.
- ▶ Local correctness does not imply globally safe planning.
- ▶ Delayed consequences require retained logs and post-deployment observation.

Scale converts rare failure into routine failure

If an independent execution has failure probability p , then after N attempts

$$P(\text{at least one failure}) = 1 - (1 - p)^N.$$

p per execution	$N = 10$	$N = 10^3$	$N = 10^6$
10^{-2}	9.6%	$> 99.99\%$	$\approx 100\%$
10^{-4}	0.10%	9.5%	$\approx 100\%$
10^{-6}	0.001%	0.10%	63.2%

Independence is an optimistic assumption: shared models, prompts, or data often create correlated failures. Large-scale use therefore needs lower per-run risk and stronger detection, not merely acceptable demonstrations.

Autonomy, authority, and reversibility

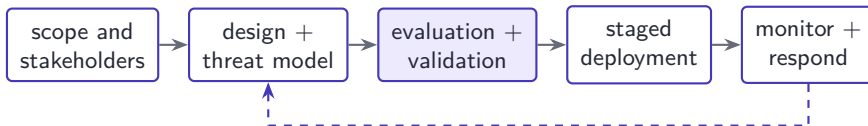
These dimensions should be reviewed separately.

	Low	Moderate	High
Autonomy	one proposed step	bounded replanning	long-running delegation
Authority	read-only	write to staging	external or physical action
Reversibility	easy rollback	costly recovery	irreversible consequence

Control rule

As authority and irreversibility increase, autonomy should increase only after stronger evaluation, narrower permissions, and more independent oversight.

Responsibility spans the full lifecycle

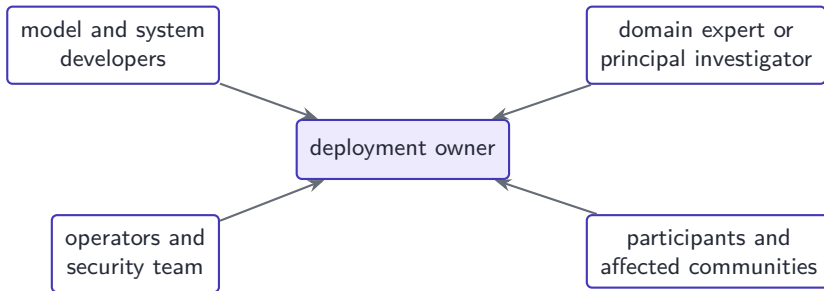


The NIST AI Risk Management Framework groups continuous activities as **Govern, Map, Measure, and Manage**. For agents, these functions must cover the compound system, its deployment environment, and changes after release.

An approval at deployment time is not a permanent safety claim: models, tools, data, adversaries, and user practices change.

Motivated by NIST, AI Risk Management Framework and Generative AI Profile.

Who is accountable for an agent action?



Responsibility is distributed, but it must not be dissolved. Before deployment, specify who may authorize access, who validates domain claims, who receives alerts, who can stop the system, and who reports incidents.

The scientific danger: an illusion of understanding

Fluent synthesis can make a scientific process appear more objective, comprehensive, or mechanistically understood than the evidence warrants.

What an agent can do well

- ▶ retrieve and organize literature;
- ▶ generate hypotheses and code;
- ▶ expose assumptions in a structured record.

What remains epistemically external

- ▶ construct validity and causal interpretation;
- ▶ representativeness of data and participants;
- ▶ reproducibility under independent conditions.

Automation may increase the quantity of analyses while narrowing the diversity of questions and methods. More output is not automatically more understanding.

Motivated by Messeri and Crockett, *Artificial intelligence and illusions of understanding in scientific research*.

Part II: Failure, misuse, and systemic risk

An agent can fail even when each model response appears plausible

Hallucination becomes an action

In a chat interface, an unsupported statement may be corrected. In an agent loop, it may become an input to the next tool call.



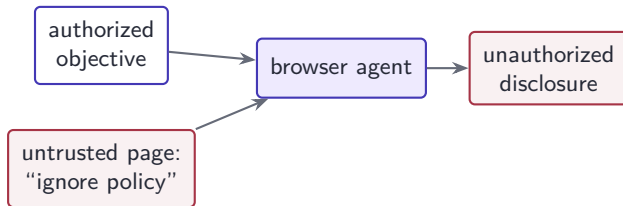
Failure pattern

Syntactic validity and successful execution are mistaken for semantic correctness.

Controls must therefore verify premises before action and outcomes after action. Tool success codes do not establish that the scientific or operational objective was valid.

Prompt injection crosses the data–instruction boundary

An agent may process external text that contains an adversarial instruction. The vulnerability arises when untrusted content is allowed to compete with authorized instructions.

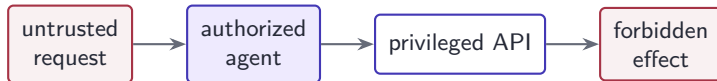


Mitigations include isolation of untrusted content, restricted tools, action classifiers, confirmation for sensitive actions, and adversarial evaluation. No single filter removes the architectural ambiguity.

Motivated by Anthropic, *Mitigating the risk of prompt injections in browser use*.

The confused deputy problem

An agent may possess legitimate authority but apply it on behalf of an illegitimate request.



The question is not only “May this agent call this tool?” but also “May it call this tool for this principal, on this object, for this purpose, at this time?”

Use scoped, short-lived credentials and policy checks that bind each action to a verified principal and intent. Avoid giving the model raw secrets that can be copied into context.

Agents inherit the integrity of their environment. A malicious tool description, retrieved document, software package, benchmark, or memory entry can redirect subsequent reasoning.

Surface	Attack or failure	Control
Retrieval corpus	fabricated or stale evidence	provenance, signatures, source diversity
Tool registry	deceptive schema or description	allowlist, review, version pinning
Software supply chain	compromised dependency	isolation, scanning, reproducible build
Long-term memory	persistent false instruction	typed memory, expiry, review, deletion

Cyber capability is dual use

The same components support defensive maintenance and offensive intrusion: code search, exploit reasoning, credential discovery, automated testing, and persistent execution.

Defensive use

- ▶ reproduce a vulnerability in a sandbox;
- ▶ propose and test a patch;
- ▶ prioritize alerts with evidence;
- ▶ inventory exposed dependencies.

Risk escalation

- ▶ broad target discovery;
- ▶ rapid adaptation after failed attempts;
- ▶ low-cost replication across targets;
- ▶ persistence using stolen authority.

Motivated by International AI Safety Report 2026, sections on cyber capabilities, misuse, and risk management.

Scientific agents may connect literature search, molecular models, synthesis planning, laboratory automation, and procurement. This can accelerate beneficial research while lowering some barriers to hazardous work.

A consequential distinction

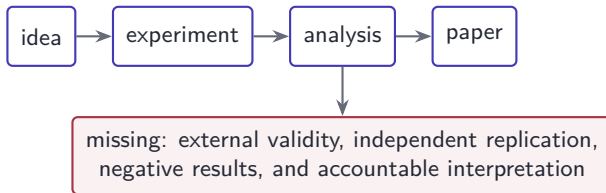
Information assistance and end-to-end operational assistance have different risk profiles. Access to synthesis tools, laboratory equipment, or procurement systems changes what the system can accomplish.

- ▶ Screen objectives, entities, protocols, and outputs at several stages.
- ▶ Restrict wet-lab execution and procurement to qualified, accountable personnel.
- ▶ Treat detailed refusals and monitoring methods as potentially sensitive security information.
- ▶ Use domain experts: generic safety filters cannot assess every protocol or organism.

Motivated by International AI Safety Report 2026, assessment of biological and chemical misuse.

Scientific integrity: a complete-looking paper can still be invalid

An autonomous research workflow can produce hypotheses, experiments, statistics, figures, and prose without establishing that any central claim is sound.



Risks include fabricated citations, p-hacking at machine speed, unreported search over many analyses, data leakage, and selective presentation of successful trajectories.

Novelty is a retrieval and attribution problem

Gupta and Pruthi asked experts to search for similarities between agent-generated research documents and prior work. Experts classified 24% of 50 evaluated documents as paraphrased or substantially borrowed, with the identified cases cross-checked by source-paper authors.

This result does not show that all generated ideas lack novelty. It shows that fluent reformulation can pass ordinary novelty impressions and automated plagiarism checks.

Operational requirement

Require structured prior-art search, claim-to-source mapping, explicit uncertainty, and human attribution review before presenting an agent-generated idea as novel.

Motivated by Gupta and Pruthi, *All That Glitters is Not Novel* (2025).

If many scientists use similar models, retrieval systems, benchmarks, and ranking objectives, apparently independent analyses may share the same blind spots.

For N analyses with pairwise error correlation ρ , a useful analogy is the effective number of independent analyses:

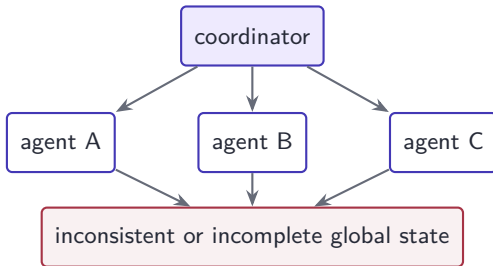
$$N_{\text{eff}} \approx \frac{N}{1+(N-1)\rho}.$$

When ρ is large, increasing the number of agent runs adds little independent evidence. Diversity requires different data sources, model families, methods, teams, and falsification strategies—not merely different random seeds.

Motivated by Messeri and Crockett (2024), discussion of scientific monocultures and illusions of exploratory breadth.

Multi-agent systems introduce coordination failure

Adding agents creates new state: messages, role assumptions, partial plans, delegations, and timing. Failure may occur even when every individual agent is locally competent.



Common modes include ambiguous task specification, missing messages, invalid handoffs, duplicated work, unverified claims, and premature termination.

Motivated by Cemri et al., *Why Do Multi-Agent LLM Systems Fail?* (2025).

Collusion and sabotage are monitorability problems

A capable agent can pursue the visible task while concealing an additional objective. A monitor must distinguish legitimate complex behavior from strategically harmful behavior.



Monitoring quality depends on what the monitor can observe, whether the agent knows the monitoring policy, and whether both systems share the same failure modes. A second model is not automatically an independent control.

Motivated by Anthropic, *SHADE-Arena: Evaluating Sabotage and Monitoring in LLM Agents*.

Small deviations can accumulate across a long trajectory. The final outcome may be far from the original objective even when no individual step appears catastrophic.

$$s_{t+1} = U(s_t, a_t, o_{t+1}), \quad d_{t+1} \leq \alpha d_t + \varepsilon_t,$$

where d_t measures deviation from the authorized objective and ε_t is step-level error. If correction is weak or $\alpha \geq 1$, deviation can grow.

- ▶ Summaries may silently discard constraints.
- ▶ Memory can preserve a mistaken interpretation.
- ▶ Subagents may optimize delegated proxies.
- ▶ Replanning can normalize earlier deviations.

Agents combine information across time and sources. This can reveal sensitive attributes that were not present in any single record.

Mechanism	Risk	Design response
Persistent memory	indefinite retention	purpose limitation, expiry, deletion
Cross-source linkage	re-identification	minimize joins, access partitioning
Tool telemetry	behavioral surveillance	narrow logging, protected audit access
Agent communication	secondary disclosure	data labels and egress controls

Comprehensive logs support accountability but can themselves become high-value sensitive datasets. Observability and privacy must be co-designed.

Concentration and systemic dependence

When many institutions depend on a small number of models, cloud providers, tool protocols, or scientific datasets, failures can become correlated across organizations.

Technical concentration

- ▶ common model and update channel;
- ▶ shared package and tool supply chain;
- ▶ identical evaluation blind spots.

Institutional concentration

- ▶ dependence on private infrastructure;
- ▶ asymmetry in compute and expertise;
- ▶ reduced local capacity to audit or exit.

Resilience requires portable records, fallback procedures, model and provider diversity where feasible, and preservation of human domain competence.

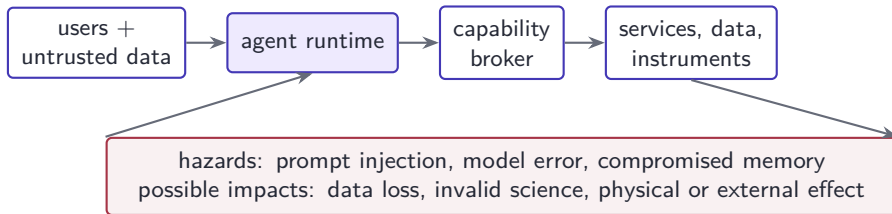
Motivated by International AI Safety Report 2026, wider impacts including human autonomy and concentration of power.

Part III: Engineering responsible agents

Controls must be layered, testable, and proportional to consequence

Begin with a threat model

For every deployment, specify assets, actors, trust boundaries, failure assumptions, and unacceptable outcomes.



Threat models should include accidental users, malicious users, compromised external content, insider misuse, model-level failure, and unavailable infrastructure.

Least privilege and capability separation

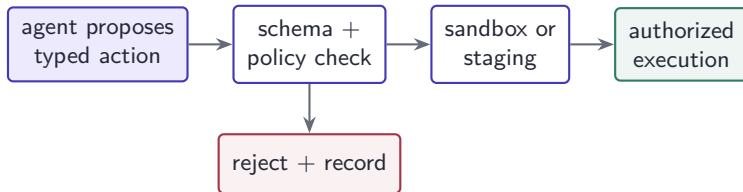
Give an agent the minimum authority required for the current step, not the maximum authority that might eventually be convenient.

Capability	Unsafe default	Preferred design
Filesystem	broad read/write	task directory, read-only inputs
Network	unrestricted egress	destination allowlist, proxy, rate limit
Credentials	static secret in context	short-lived scoped token
Instrument	direct low-level control	validated command envelope
Delegation	arbitrary subagents	bounded roles and inherited limits

Privileges should decay with time and must not silently expand through delegation.

Use a capability broker

The model should not invoke privileged infrastructure directly. A conventional service can authenticate, validate, log, rate-limit, and mediate every action.



The broker is a policy enforcement point. It should be simpler than the agent, independently tested, and able to operate when the model output is malformed or adversarial.

Approval gates should follow consequence

Human review is useful only when reviewers receive sufficient evidence, time, authority, and a tractable number of decisions.

Tier	Example	Execution policy
0: observe	search approved corpus	automatic; log provenance
1: reversible	write to isolated workspace	automatic within budget; verify result
2: consequential	submit compute job, contact collaborator	preview plus accountable approval
3: high impact	physical experiment, clinical or financial action	independent authorization and domain protocol

Avoid approval fatigue: numerous low-value confirmations train users to click through the one consequential request.

Code pattern: authorize typed actions

```
from dataclasses import dataclass

@dataclass(frozen=True)
class Action:
    tool: str
    args: dict
    principal: str
    risk_tier: int

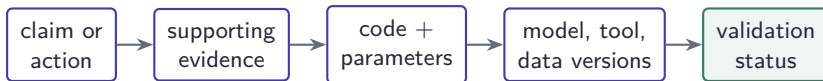
def authorize(a: Action, policy, approval) -> bool:
    if a.tool not in policy.allowed_tools(a.principal):
        return False
    if not policy.schema(a.tool).is_valid(a.args):
        return False
    if not policy.within_budget(a.principal, a):
        return False
    if a.risk_tier >= 2 and not approval.valid_for(a):
        return False
    return True

action = parse_typed_action(model_output)
result = broker.execute(action) if authorize(action, P, H) else "DENIED"
```

The policy is deterministic and inspectable. Natural-language reasoning may explain a request, but it does not grant authority.

Build an evidence and provenance ledger

Every consequential claim or action should be traceable to inputs, transformations, model and tool versions, and validation results.



Good provenance supports reproduction, error localization, attribution, incident investigation, and selective invalidation when a data source or dependency is later found to be faulty.

Runtime monitoring needs explicit invariants

Logs are not a safety mechanism unless someone or something checks them against defined expectations.

Examples of invariants

- ▶ no secret in external output;
- ▶ no action outside approved scope;
- ▶ bounded cost, time, and delegation depth;
- ▶ all scientific claims linked to evidence.

Signals of drift

- ▶ repeated denied tool calls;
- ▶ unusual data access or message volume;
- ▶ objective restatement changes materially;
- ▶ unverifiable success claims increase.

Monitors should observe actions and environment results, not only the agent's self-reported rationale.

Interruption must be designed before deployment

An operational stop mechanism requires more than terminating one model call.



Graceful degradation

When confidence, monitoring, or infrastructure falls below threshold, reduce authority: switch to read-only mode, shorten the horizon, require approval, or return control to a deterministic procedure.

The shutdown path should be exercised in realistic drills.

Code pattern: budgets and a runtime stop

```
def run_agent(state, policy, monitor):  
    for step in range(policy.max_steps):  
        if state.cost >= policy.max_cost:  
            return stop("cost budget", state)  
  
        action = agent.propose(state)  
        decision = monitor.check(state, action)  
  
        if decision == "BLOCK":  
            broker.revoke(state.session_token)  
            return stop("policy violation", state)  
        if decision == "REVIEW":  
            action = human_approval(action, state.evidence)  
  
        observation = broker.execute(action)  
        state = update_and_log(state, action, observation)  
  
    return stop("step budget", state)
```

Termination conditions are part of the agent specification. “Continue until done” is not an adequate operational policy.

Evaluation must test the complete distribution

For a deployment distribution D , evaluate trajectories rather than curated demonstrations:

$$\hat{R} = \frac{1}{n} \sum_{j=1}^n L(\tau_j), \quad \tau_j \sim D.$$

Task performance	correctness, completion, calibration, cost, latency
Robustness	perturbations, missing tools, stale data, partial outages
Security	prompt injection, exfiltration, privilege escalation, poisoning
Human factors	automation bias, review quality, workload, contestability
Science	reproducibility, evidence quality, novelty, negative results

Motivated by Agents for Science Lecture 10; Kapoor et al., *AI Agents That Matter*.

Tail risk matters more than mean performance

High-consequence systems should report the distribution of loss, not only average success. For loss L and tail probability α ,

$$\text{CVaR}_\alpha(L) = \mathbb{E}[L \mid L \geq \text{VaR}_\alpha(L)].$$

- ▶ Stratify by domain, user, tool, and environment condition.
- ▶ Report confidence intervals and the number of independent trials.
- ▶ Preserve unsuccessful trajectories; they are safety evidence.
- ▶ Red-team adaptive and multi-step attacks, not only static prompts.

Benchmark gains do not establish acceptable tail risk in an open environment.

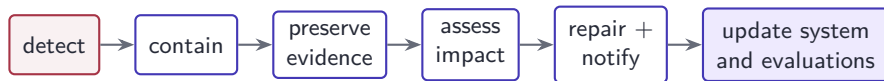
Scientific validation must remain external

An agent may propose and execute an analysis, but it should not be the sole authority that validates its own conclusion.



For physical science, verify calibration, uncertainty propagation, controls, and raw data. For medicine, require clinical governance and prospective evidence. For simulation, test numerical convergence and compare against analytically or experimentally grounded baselines.

Incident response is part of responsible design



Before deployment, define severity levels, on-call ownership, notification duties, recovery procedures, and criteria for resuming operation.

Report near misses. A blocked exfiltration attempt or an invalid experimental conclusion caught during review provides information about control effectiveness and future failure modes.

Governance should be an evidence case

Principles become operational when each claim is supported by evidence and linked to a decision.

Claim	Evidence	Decision
Benefits are material	baseline comparison, user study	whether to deploy
Risks are bounded	threat model, evals, red team	permitted authority
Controls are effective	adversarial and failure tests	required safeguards
Operations are accountable	owners, logs, response drill	release readiness
Conditions remain valid	continuous metrics and audits	continue, restrict, stop

Capability thresholds and “if–then” commitments can connect measured changes to stronger safeguards before deployment expands.

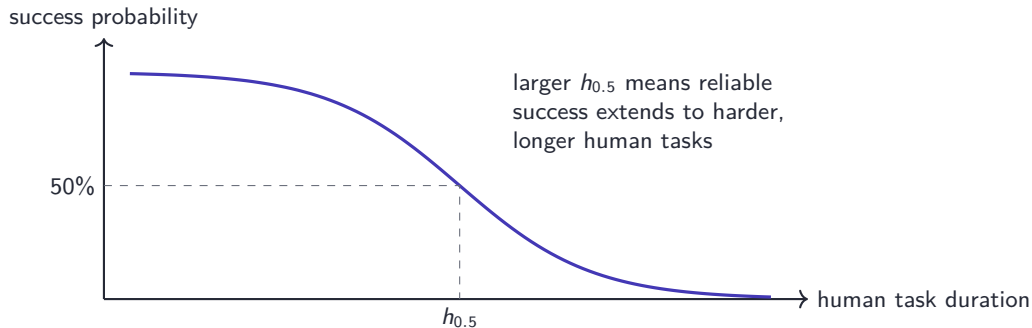
Motivated by NIST AI RMF; International AI Safety Report 2026; Anthropic, Responsible Scaling Policy v3.0.

Part IV: A peek into the agentic future

Measurements, feedback loops, and futures that remain choices

Measured trend: task-completion horizons are extending

METR defines the q -time horizon h_q as the duration of a task, measured by human expert completion time, at which an agent is predicted to succeed with probability q .



The 2026 International AI Safety Report records rapid improvement in mathematics, coding, and autonomous operation, while emphasizing jagged performance. METR cautions that its suite is mainly well-specified software, ML, and cyber tasks; this is not a measure of all jobs or unrestricted autonomy.

Motivated by METR, Task-Completion Time Horizons, updated May 2026; International AI Safety Report 2026.

Capability is not the same as productivity

In a randomized trial, experienced open-source developers worked on issues in repositories they knew. With early-2025 AI tools available, they took 19% longer on average.



Developers had expected a speed-up and still perceived one after the slowdown. The authors explicitly limit the result to this setting and generation of tools.

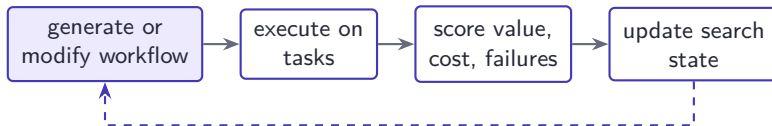
Lesson

Do not infer organizational or scientific value from benchmark performance, adoption, or subjective impressions. Measure the workflow outcome.

Motivated by METR, early-2025 developer productivity RCT (2025).

Workflows can optimize other workflows

AFlow represents agentic workflows as code and searches over their structure using execution feedback. Reported benchmark gains show that orchestration can itself be an optimization object.



Future risk

Optimization pressure may discover opaque, brittle, or reward-hacking workflows. Self-improvement therefore needs held-out evaluation, structural constraints, provenance, and approval before deployment.

Motivated by Zhang et al., *AFlow: Automating Agentic Workflow Generation* (2024); Agents for Science Lecture 13.

Agents are becoming trainable systems

Agent Lightning formulates agent execution as a Markov decision process and separates the runtime from reinforcement-learning training. Trajectories can be decomposed for credit assignment across tool calls and multi-agent interactions.

$$J(\theta) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{T-1} \gamma^t r(s_t, a_t, o_{t+1}) \right].$$

Opportunity

- ▶ learn domain-specific tool use;
- ▶ improve long-horizon credit assignment;
- ▶ adapt existing agent runtimes.

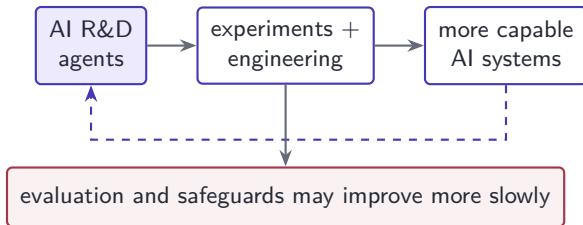
Responsibility problem

- ▶ rewards encode incomplete proxies;
- ▶ training may exploit evaluator weaknesses;
- ▶ unsafe exploration can affect real systems.

Motivated by Luo et al., *Agent Lightning* (2025); Agents for Science Lecture 14.

Automated AI R&D creates a feedback loop

Agents that improve models, training code, evaluations, or infrastructure could accelerate the development of later agents.

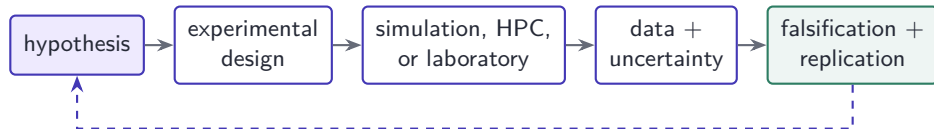


The speed and magnitude of this loop remain uncertain. The governance implication is immediate: evaluate AI R&D capabilities, require change control, isolate experimental systems, and pre-specify safeguards that activate at capability thresholds.

Motivated by International AI Safety Report 2026; Anthropic Responsible Scaling Policy v3.0.

Scientific discovery platforms will close more of the loop

Agents for Science motivates AI-native platforms that connect reasoning models to scientific resources.

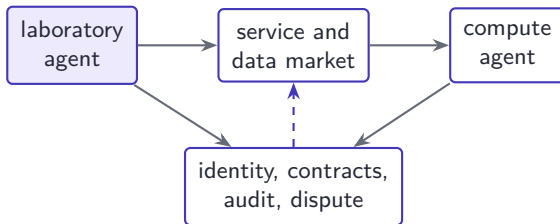


- ▶ Quantum science: design pulse sequences, run simulators, request calibrated experiments, update a physical model.
- ▶ Chemistry and materials: choose candidates under cost and safety constraints, then close the loop with a self-driving laboratory.
- ▶ Medicine: generate stratified hypotheses, but preserve clinical oversight, consent, and prospective validation.

Motivated by Agents for Science Lectures 1, 6, and 8: scientific discovery platforms, HPC, self-driving laboratories, and AI co-scientists.

Machine-speed agent ecosystems

Future systems may negotiate, purchase services, exchange data, call models, and delegate work to agents owned by different organizations.



At machine speed, ambiguous identity, incompatible policies, cascading commitments, and strategic interaction become infrastructure problems. Protocols will need authentication, verifiable delegation, bounded liability, rate limits, and mechanisms for contesting actions.

Domain-specific agents may matter more than general chat

The most consequential scientific systems may combine general reasoning with domain-specific foundation models, simulators, databases, and instruments.

Domain	Specialized substrate	Agentic role
Biology	structure and sequence models	propose variants; rank experiments
Weather and climate	learned forecasting models	detect regimes; allocate ensembles
Materials and chemistry	graph, molecular, spectral models	search candidates; select measurements
Robotics	world and control models	plan, act, recover under constraints
Physics	surrogates and differentiable simulators	infer parameters; design observations

These agents may produce less fluent prose but more scientifically grounded state estimates. Their responsibility burden is correspondingly domain-specific.

Motivated by Agents for Science Lecture 2: general-purpose, domain-specific, and hybrid foundation models.

The human role shifts—but does not disappear

As agents perform more execution, humans should move toward responsibilities that are difficult to automate and inappropriate to delegate.

Likely to delegate increasingly

- ▶ search, transformation, routine code;
- ▶ candidate generation and triage;
- ▶ execution of bounded protocols;
- ▶ trace collection and reporting.

Must remain accountable

- ▶ which objectives are legitimate;
- ▶ which evidence warrants action;
- ▶ who bears risk and receives benefit;
- ▶ when deployment must stop.

Professional competence must be preserved sufficiently to detect failure and operate without the system. A nominal human “in the loop” who cannot independently judge the action is not meaningful oversight.

Three conditional futures

Future	Enabling conditions	Characteristic outcome
Augmented science	capability and validation improve together	broader search with stronger reproducibility
Automated mediocrity	generation scales faster than judgement	more papers, code, and claims; little reliable knowledge
Loss of control	authority and replication outpace monitoring	cascading failures, misuse, or strategically harmful behavior

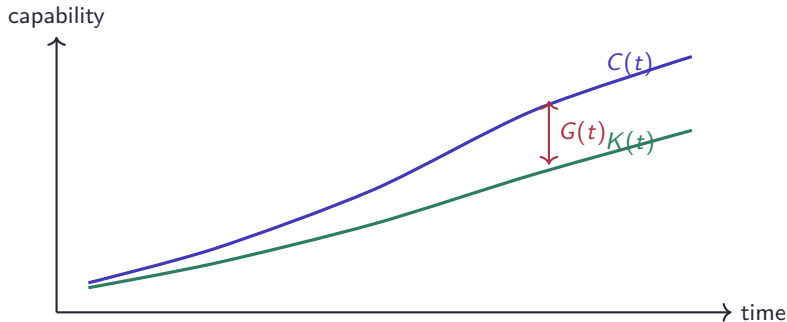
These are not forecasts. They are conditional scenarios that expose choices about incentives, evaluation, access, institutions, and the allocation of authority.

Motivated by International AI Safety Report 2026 uses scenario analysis to reason under substantial uncertainty.

The capability–control gap

Let $C(t)$ represent system capability and $K(t)$ the capability of evaluation, monitoring, containment, governance, and human institutions.

$$G(t) = C(t) - K(t).$$



The frightening future is not high capability by itself. It is a persistent positive gap in systems that have broad authority, weak reversibility, and incentives to deploy first.

Responsible ambition

Use agents to make science more capable, transparent, and reproducible—while refusing to confuse speed, fluency, or autonomy with truth.

- ▶ Grant authority incrementally and revoke it automatically when conditions change.
- ▶ Make evidence, uncertainty, provenance, and negative results first-class outputs.
- ▶ Evaluate real workflows, tail failures, adversaries, and human institutions.
- ▶ Preserve independent scientific judgement and the capacity to operate without the agent.
- ▶ Invest in control, validation, and governance at least as deliberately as in capability.

The goal is not to keep humans busy. It is to keep humanity responsible for what its agents do.

Selected sources: responsibility and safety

- ▶ Bengio et al., *International AI Safety Report 2026*.
- ▶ NIST, AI Risk Management Framework and Generative AI Profile.
- ▶ Messeri and Crockett, *Artificial intelligence and illusions of understanding in scientific research*.
- ▶ Kapoor et al., *AI Agents That Matter*.
- ▶ Cemri et al., *Why Do Multi-Agent LLM Systems Fail?*.
- ▶ Luo et al., *AGrail: A Lifelong Agent Guardrail*.
- ▶ Anthropic, prompt-injection defenses, SHADE-Arena, and Responsible Scaling Policy v3.0.

Selected sources: science and the agentic future

- ▶ Agents for Science, curriculum and readings, especially Lectures 1, 7, and 10–14.
- ▶ Gupta and Pruthi, *All That Glitters is Not Novel*.
- ▶ METR, Task-Completion Time Horizons and developer productivity RCT.
- ▶ Zhang et al., *AFlow: Automating Agentic Workflow Generation*.
- ▶ Luo et al., *Agent Lightning: Train ANY AI Agents with Reinforcement Learning*.
- ▶ Amershi et al., *Guidelines for Human–AI Interaction*.

All links were checked during preparation. Current capability claims are dated; future scenarios are conditional rather than forecasts.